Jonghee Jeon

Ph.D. candidate · NMAIL, School of Computing, KAIST
Advisor: Prof. Sungho Jo

jjeon365@kaist.ac.kr
kaist-jonghee.github.io
github.com/kaist-jonghee · ORCID 0009-0009-5508-8170

Our bodies know things before we do. I use computer science to read their signals and bring that knowledge back to people.

EDUCATION

Sep 2025–now — **Ph.D. candidate in Computer Science**, KAIST School of Computing
Advisor: Prof. Sungho Jo (NMAIL)

Mar 2023–Aug 2025 — **M.S. in Computer Science**, KAIST School of Computing
Thesis: *Design and Behavioral Segmentation of an IMU-Based Wearable for Canine Health Monitoring*

Mar 2019–Feb 2023 — **B.S. in Computer Science, minor in Biology**, KAIST
Thesis: *Development of a Smart-Earphone that plays music according to the user's mental state*

PUBLICATIONS

2026 — **High-Field-Gradient Nanoplasmonic Platform Enables Small-Molecule Dielectrophoresis and SERS Detection.**
Minjoon Kim†, Sejoon Huh†, Minjae Ku, Seokjun Cha, Gunwoo Jo, **Jonghee Jeon**, Seunghee H. Cho, Byeong-Ho Jeong, In Woong Han, Min-Young Lee*, Sungho Jo*, Yeon Sik Jung*. *Advanced Functional Materials*, accepted. († equal contribution, * corresponding)

2022 — **Enhancing the performance of P300-based BCIs by tDCS of the left VL-PFC.**
International Winter Conference on Brain–Computer Interface (BCI 2022).

RESEARCH

2025–now — **Explaining biological Raman/SERS spectra** (Ph.D.)
Assigning each band of a measured spectrum to a molecule, a vibration and its environment, so a spectrum counts as explained when the residual matches replicate noise. Studies on peptides, DNA methylation and urine.

2022–2025 — **Automatic Raman preprocessing** (with CJ)
Deep-learning regression and automatic preprocessing search that lets PLS and other chemometric models beat raw PLS.

2023–2025 — **Smart collar for canine health** (M.S.)
Collar IMU with semi-automatic video labelling and segmentation of 20 dog behaviours, daily activity reports, pedometer and gait features.

2024 — **Full-body motion from fewer IMUs; SLoMo**
Pose reconstruction from 4–5 IMUs with transformers; sports location from a single wrist IMU.

2020–2022 — **Wearable physiological sensing**
In-ear PPG with motion-artifact removal and HRV-based emotion classification; PPG sensing pencil; dog heart sensing through fur with PPG and a silicone stethoscope.

EXPERIENCE

May 2024–Aug 2026 — **Co-founder and CEO, rAI** · rai-sw.com
Founded and led the company; built rAI-Q, Raman and SERS analysis software for non-specialists.

2023	**Semiconductor anomaly detection, Samsung Electronics** Classification models and signal data augmentation for high-dimensional process time series.
2022–2025	**Raman spectroscopy project, CJ** Regression on undisclosed targets, then automatic preprocessing for chemometric models.
Nov 2020–Feb 2023	**Undergraduate researcher, NMAIL, KAIST** Wearable PPG, acoustic and IMU sensing.

TEACHING AND MENTORING

Fall 2026	**TA, CS572 Intelligent Robotics**, KAIST
Spring 2026	**TA, CS270 Intelligent Robot Design and Programming**, KAIST
Fall 2025	**TA, CS572 Intelligent Robotics**, KAIST
Fall 2023	**TA, COE202 Basics of Artificial Intelligence: Physical AI**, KAIST · **Outstanding TA Award**
Spring 2023	**TA, CS270 Intelligent Robot Design and Programming**, KAIST · **Outstanding TA Award**
Mar–Nov 2024	**Research advisor, KAIST HRP** · mentored Yernar Amanbek on full-body motion from fewer IMUs
Jul–Dec 2023	**Research mentor, KAIST URP** · mentored 이제형